LearnSync AI: An Artificial Intelligence-Based Personalized Learning Companion for Pasig Christian Academy

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CHAPTER I

THE PROBLEM AND ITS BACKGROUND

The rapid growth of digital learning has made independent study an important part of modern education. Students increasingly use digital learning applications, such as review tools, flashcards, and quiz platforms, to support lessons outside the classroom. However, many existing educational platforms still rely on a one-size-fits-all learning approach, where the same learning activities and instructional materials are provided to all learners regardless of their individual learning progress or needs. Recent literature identifies personalized learning as a shift away from this traditional approach toward adaptive, learner-centered instruction (Fariani et al., 2023; Wang et al., 2025).

As a result, students may spend too much time reviewing concepts they already understand while receiving limited support for topics they find difficult. Learning activities that do not match students' knowledge and skill levels may also reduce engagement and contribute to boredom during independent study (Krannich et al., 2025). These challenges highlight the need for more personalized learning tools that can better address students' individual learning needs.

Cognitive Load Theory suggests that learning is more effective when instructional materials are designed to align with the learner’s limited working-memory capacity. In digital learning environments, excessive text, poorly

integrated multimedia, and distracting interface elements can impose irrelevant cognitive load, which reduces learning efficiency and weakens knowledge achievement (Surbakti et al., 2024; Krieglstein et al., 2023). Because learners differ in prior knowledge, attention, language proficiency, and learning ability, one-size-fits-all platforms are less effective than adaptive systems that create content, regulate pacing, and tailor presentation to individual needs (Surbakti et al., 2024).

Similarly, digital learning environments that adapt to students' learning needs provide more effective and engaging learning experiences by reducing unrelated cognitive load (Skulmowski & Xu, 2022). These findings highlight the importance of designing educational technologies that personalize learning according to students' knowledge and learning progress.

Digital learning technologies have improved opportunities for independent learning by giving students flexible access to educational resources anytime and anywhere. This flexibility allows learners to study at their own pace, adapt learning to their individual needs, and integrate education more easily into daily life (Sisouvong & Pasanchay, 2024; Josué et al., 2023). Digital platforms also provide learning tools such as quizzes, assignments, interactive resources, discussion spaces, and feedback systems that support active participation and continued engagement with course content (Alshamary & Alhalafawy, 2023; Josué et al., 2023).

However, many existing educational applications focus on only specific learning activities instead of providing a complete learning environment. Therefore, students may rely on several non-integrated applications to support different learning needs. Research has shown that this fragmented use of multiple learning platforms can increase unrelated cognitive load and tool fatigue, highlighting the importance of more integrated digital learning environments (Qiu & Qiu, 2026).

Integrating learning activities and learning analytics within a single digital platform provides a more effective learning experience and better supports students' self-regulated learning because it connects course participation, progress monitoring, reflection, and feedback in one environment. Studies show that platforms that embed self-regulated learning supports into digital learning systems improve students' achievement, digital competencies, self-regulation, and engagement compared with conventional platform use or traditional instruction (Elmabaredy & Gencel, 2024; Yang et al., 2024; Du et al., 2025). Learning analytics strengthens this effect by helping students observe their progress, identify weaknesses, plan next steps, and adjust learning strategies, especially when analytics tools are integrated with instruction and guidance rather than used as stand-alone tracking systems (Kleimola et al., 2024, Du et al., 2025).

Artificial Intelligence (AI) has become a promising technology for improving digital learning platforms by providing personalized and adaptive learning experiences. Unlike traditional educational applications that use fixed

learning activities, AI-powered systems can analyze students' learning progress, identify their strengths and weaknesses, and provide personalized learning paths, instructional content, and feedback based on their individual learning needs (Barrera Castro et al., 2024).

Recent research shows that AI-driven adaptive learning technologies improve learning experiences by adjusting instructional support based on students' performance and learning progress. This allows students to learn at their own pace while receiving personalized guidance (Bayly-Castaneda et al., 2024). These capabilities enable digital learning platforms to provide continuous and personalized support throughout the learning process, making learning more adaptive to each student's needs.

Pasig Christian Academy (PCA) is a private educational institution established in 1986 that offers Preschool, Elementary, and Junior High School education in Pasig City. Operating within the basic education sector, the institution is committed to fostering academic excellence and preparing its learners for higher education. This study specifically focuses on the Junior High School (JHS) department, where students gradually develop independent learning habits in preparation for the academic demands of Senior High School.

Based on the researchers' requirement analysis and consultation with the school administration, independent learning at Pasig Christian Academy primarily relies on official textbooks and teacher-provided instructional materials. These learning resources remain the primary references used by students during

independent study and effectively support the school's curriculum. However, these resources are primarily static, providing the same learning content and review materials to all students regardless of their individual mastery levels, learning pace, or specific learning needs. Furthermore, the researchers' needs assessment revealed that students experience difficulty identifying which topics require additional review, desire more engaging study activities, value continuous progress monitoring, and prefer a single platform that integrates multiple independent learning functions. These findings indicate opportunities to enhance the school's existing independent learning process through more personalized, adaptive, and engaging learning support.

Rather than replacing the school's existing instructional practices, these findings present an opportunity to enhance the current independent learning process through educational technology. LearnSync AI is proposed as an independent learning platform designed to complement the school's existing learning resources by providing personalized learning support, adaptive learning activities, AI-assisted study strategies, gamified learning activities, and learning analytics to support independent learning. The platform integrates Bayesian Knowledge Tracing (BKT) to estimate each student's knowledge mastery and Generative Artificial Intelligence (Generative AI) to generate personalized learning support based exclusively on educator-uploaded instructional materials. To maintain curriculum alignment and academic accuracy, only authenticated educators are authorized to upload learning resources and assign the

appropriate grade levels that may access them, while students receive AI-generated learning support derived solely from these verified materials.

LearnSync AI integrates several evidence-based study strategies within a single platform, including Adaptive Flashcards, Adaptive Test Mode, AI-assisted study strategies such as the Feynman Technique and Free Recall, an AI-powered learning assistant, AI-generated educational games, and learning analytics. Bayesian Knowledge Tracing (BKT) continuously estimates students' mastery levels based on their assessment performance, allowing Adaptive Flashcards and Adaptive Test Mode to adapt review activities according to each learner's estimated understanding. AI-generated educational games utilize these mastery estimates to tailor question difficulty, while the AI-assisted study strategies provide individualized feedback based on students' responses. Students also receive personalized learning analytics through dashboards that display mastery progression, study streaks, total study hours, digital badges, and performance statistics, encouraging greater self-awareness and independent learning.

Unlike conventional review materials that provide the same learning experience to every learner, LearnSync AI complements the school's existing independent learning process by integrating adaptive learning activities, AI-assisted personalized feedback, evidence-based study strategies, gamified learning activities, and learning analytics into a single platform. By combining educator-approved learning resources with Bayesian Knowledge Tracing (BKT) and Generative Artificial Intelligence (Generative AI), the proposed platform aims

to provide a more personalized, engaging, and organized independent learning experience while preserving the school's existing curriculum and instructional practices.

REVIEW OF RELATED LITERATURE

This chapter presents a comprehensive synthesis of local and international literature, research studies, and technological frameworks relevant to the development of LearnSync AI.

Modern Study Challenges and Diverse Student Learners

Developing an effective digital learning platform requires recognizing that students differ in how they learn, process information, and respond to instruction. Learners vary in their prior knowledge, academic abilities, motivation, learning pace, and study strategies, so a one-size-fits-all approach does not effectively support all learners (Castro et al., 2024; Gligorea et al., 2023). Research on personalized education suggests that learning environments become more effective when they consider individual learner characteristics and provide appropriate support (Castro et al., 2024). Likewise, adaptive learning has been identified as an alternative to traditional one-size-fits-all learning approaches

because it provides learning experiences that respond to individual learner needs and progress (du Plooy et al., 2024; Gligorea et al., 2023).

In addition, studies on adaptive learning emphasize that educational systems that respond to learners' needs and progress can create more engaging and effective learning experiences (du Plooy et al., 2024; Gligorea et al., 2023). As digital learning continues to expand, educational platforms are increasingly expected to provide flexible and personalized learning experiences that address the diverse needs of students (Castro et al., 2024).

Fast and Slow Learners

Students differ in the speed at which they process information, understand new concepts, and complete learning activities. These differences influence how learners respond to classroom instruction and digital learning environments. In mixed-ability classrooms, some students quickly master lessons and may become less engaged when instruction progresses too slowly, while others require additional time, practice, and guidance before fully understanding the material. Because learners have different academic needs, using the same teaching pace for everyone

may reduce engagement and limit learning opportunities for both high-performing and struggling students. Research on differentiated instruction suggests that learning environments should provide flexible support and varied learning opportunities to accommodate differences in students' abilities and learning pace (Ziernwald et al., 2022).

Likewise, research on educational differentiation indicates that adapting learning experiences to individual learner needs may improve participation, motivation, and academic achievement in diverse classrooms (Eikeland & Ohna, 2022). These findings highlight the importance of designing learning platforms that can support students at different levels of understanding rather than requiring all learners to progress at the same pace.

Digital Overload

The increasing use of digital technologies in education has created learning environments in which students often depend on multiple digital tools and platforms to access learning materials, complete assessments, submit assignments, and communicate with instructors and peers. Although these technologies can improve access and

flexibility, managing different interfaces and constantly navigating between separate systems can increase the mental effort required to complete learning tasks. This added effort can reduce the amount of cognitive capacity available for understanding, processing, and retaining new information, especially when students must divide their attention between learning content and platform navigation (Qiu & Qiu, 2026; Ibrahim et al., 2025).

Research on digital learning further suggests that unnecessary complexity in online environments contributes to extraneous cognitive load, meaning mental effort caused by the design of the learning system rather than by the learning material itself, which can reduce learning efficiency and make it more difficult for students to focus on instructional content (Nordin et al., 2026).

Furthermore, studies in learning analytics emphasize the importance of designing organized and accessible digital learning environments that support students through efficient access to educational resources and activities while minimizing unnecessary barriers to learning (Qiu & Qiu, 2026). Taken together, these findings highlight the value of developing integrated learning platforms that centralize academic resources and learning activities, thereby creating

a more organized, efficient, and student-centered learning experience.

Junior High School Students

Digital learning in the Philippines has increasingly required students to study with greater independence, especially in online and modular environments where teacher supervision is reduced and learners must manage their own time, tasks, and study strategies. Research shows that self-regulated learning is important in these settings because students differ in their ability to set goals, structure their learning environment, seek help, and monitor their progress, and weaker self-regulation is associated with poorer learning outcomes (Funa et al., 2023; Briones et al., 2023; Calamlam, 2023). At the same time, many Filipino learners continue to experience barriers such as limited internet connectivity, inadequate resources, difficulty understanding modules, and fragmented learning conditions, which can make independent learning more challenging (Bustillo & Aguilos, 2022; Agayon et al., 2022; Abejuela et al., 2023).

These conditions support the need for learning platforms that provide organized access to resources, adaptive support, and timely feedback. Studies on educational technology further suggest that students' learning experiences improve when digital systems are designed to be accessible, interactive, well-facilitated, and responsive to learner needs, including the use of adaptive learning systems and other technology-based interventions that support engagement and performance (Batucan et al., 2022; Llacuna et al., 2026; Caliston, 2025).

Thus, developing an integrated and adaptive learning platform for Junior High School students is well grounded in the literature, as this population is still strengthening independent learning habits and can benefit from personalized support, efficient resource access, and technology that aligns with their developmental and educational needs (Salva et al., 2022; Villarama et al., 2022; Castro et al., 2024).

The reviewed literature shows that students have different learning abilities, learning pace, and study habits. Because of these differences, a one-size-fits-all approach may not effectively support every learner.

Studies suggest that personalized and adaptive learning can better address individual learning needs by providing learning experiences that match each student's progress and level of understanding.

The literature also shows that students often rely on multiple digital tools and learning resources, which can make studying more difficult due to fragmented information and increased cognitive load. These findings highlight the need for a centralized learning platform that organizes learning resources and provides a more efficient and focused learning experience.

In addition, Junior High School students are still developing independent learning and self-regulated learning skills. Many learners experience challenges such as identifying their knowledge gaps, monitoring their learning progress, and maintaining effective study habits. Therefore, they can benefit from learning technologies that provide personalized guidance, adaptive learning activities, and timely feedback.

Overall, the reviewed literature supports the development of LearnSync AI as an AI-powered adaptive learning platform for Junior High School students. By integrating Generative Artificial Intelligence (GenAI) and Bayesian Knowledge Tracing (BKT), the platform aims to provide personalized learning support, centralized learning resources, and learning analytics to help students develop more effective independent learning habits.

Analysis of Independent Study Tools and Research Gaps

Independent study tools have become an important part of modern education because they give students flexible ways to review lessons and learn outside the classroom. However, many of these tools support only specific learning activities, which can lead students to rely on multiple applications and create a fragmented digital learning experience. Evidence shows that multi-platform learning environments increase extraneous cognitive load and tool fatigue, reducing students' ability to focus on actual learning tasks rather than on managing different technologies (Qiu & Qiu, 2026; Ibrahim et al., 2025).

Research also shows that structured instructional design and interactive learning environments can reduce cognitive load, suggesting that more integrated digital platforms would better support independent learning than disconnected single-purpose tools (Nordin et al., 2026). These findings reveal an important research gap in existing independent study tools, particularly the limited evidence on platforms that combine multiple evidence-based learning functions with adaptive support in a single, cognitively manageable environment.

Fragmentation of Current Educational Applications

Educational applications have become widely used to support independent learning by offering tools such as flashcards, quizzes, note-taking, and progress tracking. However, many of these applications are designed to support only specific learning tasks, so students often need to move across several platforms to complete different parts of their study routine. This creates a fragmented digital learning experience in which attention is divided between learning content and managing multiple technologies. Recent research shows that multi-platform learning environments increase extraneous cognitive load, contribute to tool fatigue, and make it harder for students to focus on meaningful learning tasks rather than navigation, logins, and platform switching (Qiu & Qiu, 2026; Ibrahim et al., 2025).

Research also suggests that fragmented learning resources often lack coherent organization, which further reduces learning efficiency and makes knowledge construction more difficult (Hu & Liu, 2025). Supporting this, integrated resource designs have been associated with better perceived coherence and lower cognitive load,

indicating that more unified educational systems may better support independent learning than disconnected single-purpose applications (Lou, 2025).

Limitations of Static Review Systems

Many educational applications still rely on static review systems that give all learners the same materials and review activities, even though learners differ in pace, prior knowledge, and current understanding (Raj & Renumol, 2022; Bakri, 2025). This one-size-fits-all structure can be inefficient because conventional review often distributes the same materials uniformly rather than targeting each learner's actual weaknesses (Raj & Renumol, 2022; Okubo et al., 2023) Research on personalized learning instead describes adaptive systems as ones that adjust to changing learner needs and recommend resources based on learner characteristics, knowledge level, and preferences (Raj & Renumol, 2022).

Evidence also indicates that adaptive review can improve learning: a personalized review dashboard that recommended insufficiently understood pages produced a significantly higher improvement in quiz scores, while a

broader review found that adaptive systems often improve academic performance and engagement compared with non-adaptive environments (Okubo et al., 2023; du Plooy et al., 2024; Bakri, 2025).

The reviewed literature shows that many existing educational applications either support only specific learning activities or continue to rely on one-size-fits-all review approaches. As a result, students may experience fragmented learning environments and limited opportunities for personalized learning support. This fragmented approach can increase cognitive load, reduce learning efficiency, and make it more difficult for students to focus on understanding lessons. In addition, many current review systems continue to use a one-size-fits-all approach, providing the same learning materials and activities to all students despite differences in learning pace, prior knowledge, and individual learning needs.

Although existing applications offer useful learning features, the literature indicates a need for a more integrated and adaptive learning platform that combines personalized learning, organized learning resources, and continuous learning support within a single environment. Guided by these findings, LearnSync AI is proposed to address these research gaps by integrating Generative Artificial Intelligence (GenAI), Bayesian Knowledge Tracing (BKT), adaptive learning activities,

AI-assisted learning support, gamification, and learning analytics into one centralized platform. Through these features, the platform aims to provide Junior High School students with a more personalized, organized, and engaging independent learning experience.

Learning Analytics and Student Motivation

Learning analytics has become an important educational tool because learner-facing dashboards give students direct access to information about their progress, helping them monitor performance and identify what they still need to study. Unlike static review systems that provide the same materials to all learners, dashboards can support self-regulated learning by making progress visible and guiding reflection, feedback, and next actions (Paulsen & Lindsay, 2024; Yang et al., 2024). Experimental and quasi-experimental evidence suggests that these systems can improve students' awareness of their learning, self-efficacy, engagement, and academic outcomes when the dashboards are intentionally designed to support motivation and regulation rather than simply display data (Fleur et al., 2023; Yang et al., 2024).

Research also indicates that dashboards work best when grounded in motivational theory and designed to help learners make autonomous, strategic decisions, highlighting the limitations

of static one-size-fits-all review approaches that do not adapt to students' ongoing progress (Lee & Kim, 2024;

Mastery Progress

Visualizing mastery progress helps students engage in self-monitoring because it makes their current standing and learning trajectory visible during the learning process, not only after formal grading events (Paulsen . In competency-based settings, progress monitoring is tied to feedback, reflection, and self-regulation, with regular feedback helping students understand achievement and adjust strategies to reach learning goals

Learner-facing dashboards have increasingly shifted from teacher-oriented analytics tools to student-focused systems intended to support learning directly. A systematic review found that monitoring, awareness, and comparison are the most common dashboard affordances, but more recent designs increasingly include feedback, reflection, and goal-setting features that better support learning action

Evidence from higher education also suggests that dashboards can increase engagement with self-assessment systems. In a study of 194 students, those with dashboard

access interacted more with a computerized classification testing system and reported higher confidence than students using the same system without a dashboard. The dashboard in that study displayed mastery-related indicators such as the number of mastered subject areas, correct and incorrect responses, and total attempts, showing how progress visualizations can make formative assessment more usable for students across a semester

Study Consistency and Behavioral Engagement

Study streaks and digital badges are features that encourage students to develop consistent learning habits by recognizing their regular participation and achievements. A study streak records the number of consecutive days a student completes learning activities, while digital badges acknowledge milestones such as completing lessons or reaching learning goals.These features provide continuous feedback and recognition that can motivate students to remain engaged and participate consistently in learning activities (Sailer & Homner, 2020).

Research has shown that consistent engagement and self-monitoring help students develop self-regulated learning

skills and improve their persistence in completing learning activities (Matcha et al., 2020). Learning analytics studies also indicate that monitoring students' behavioral engagement helps promote active participation and supports improved learning performance by encouraging consistent involvement in learning activities (Johar et al., 2023).

In addition, achievement indicators such as digital badges have been found to increase learner engagement and encourage continued participation when they are linked to meaningful learning goals and progress (Tahir et al., 2022). Collectively, these findings suggest that study streaks and digital badges can promote consistent learning habits, strengthen learner engagement, and support regular participation in independent learning.

Time-on-Task and Self-Regulated Learning

Time-on-task refers to the amount of time students spend completing learning activities. In digital learning environments, tracking and displaying study time helps students monitor their learning habits and become more aware of how they manage their time. Unlike traditional learning environments, where students' independent study

time is often difficult to monitor, digital platforms provide real-time information that allows learners to evaluate their study pace and identify whether they are spending enough time on their lessons (Hilliger et al., 2021).

Research has shown that monitoring learning behaviors, including time spent on learning activities, supports self-regulated learning by encouraging students to plan, monitor, and reflect on their learning process (Lim et al., 2021). In addition, learning analytics that visualize study behaviors help students become more aware of their learning progress and encourage them to take greater responsibility for their own learning (Aguilar et al., 2021). These findings suggest that displaying time-on-task information can promote greater responsibility for learning, support independent learning, and encourage students to develop effective study habits.

The reviewed literature demonstrates that providing students with direct access to their own learning data supports motivation and self-regulated learning by helping learners monitor their progress and study behaviors. Features such as visualizing mastery progress, tracking study time, and using gamified rewards like streaks and badges help

learners build consistent study habits and monitor their own performance. These tools are especially important for students who need to manage their time and identify their own weaknesses during unsupervised study.

Guided by these findings, LearnSync AI features a personalized learning analytics dashboard designed to support Grade 10 students during their independent studies. The platform centralizes their educational data by displaying real-time mastery progress, total study hours, study streaks, and digital badges. By transforming students' learning activities into clear, visual feedback, the platform encourages them to take greater responsibility for their own learning, maintain consistent study routines, and actively monitor their academic progress without requiring continuous teacher supervision.

Using Artificial Intelligence for Personalized Student Support

Artificial Intelligence has become an important component of modern educational systems by providing personalized support that addresses the diverse learning needs of students. Traditional learning approaches often rely on fixed instructional materials that do not adapt to individual differences in learning pace, prior knowledge, or performance. In contrast, AI-based educational systems can analyze student performance and learning behaviors to personalize instructional content, provide timely feedback, and

recommend learning activities based on individual progress (Chen et al., 2020). By adapting learning experiences to each student's needs, AI has the potential to improve student engagement, promote deeper understanding, and support more effective learning outcomes.

Beyond personalizing learning content, AI can also provide continuous academic support by offering personalized feedback, recommending appropriate learning activities, and identifying areas where students need additional practice or guidance. Unlike traditional educational platforms that present the same content to all learners, AI-powered systems can respond to individual learning needs by adjusting learning support based on students' progress and performance. This personalized approach enables students to learn at their own pace while receiving timely assistance that strengthens their understanding and encourages continuous learning (Chen et al., 2020).

AI in Modern Classrooms

The integration of Artificial Intelligence in modern classrooms has played an important role in shifting traditional educational approaches toward more flexible and adaptive learning environments. Historically, many learning

systems have followed a one-size-fits-all approach in which students receive the same lessons and review materials regardless of their individual abilities, learning pace, or level of understanding. In contrast, AI-based educational systems can analyze student interactions and performance data to personalize instructional content according to individual learning needs (Chen et al., 2020).

These systems provide personalized feedback, adjust task difficulty, and recommend learning activities based on each student's current level of understanding, allowing learners to receive appropriate support throughout the learning process (Murtaza et al., 2022). This adaptive capability enables students to learn at their own pace while improving engagement, learning efficiency, and independent study.

AI in Finding Student Learning Gaps

Artificial Intelligence has become an important tool for identifying students' learning gaps and providing personalized academic support. Traditional assessment methods often rely on periodic tests and evaluations, which may not immediately identify areas where students are

struggling. In contrast, AI-driven adaptive learning systems can continuously analyze students' learning behaviors and performance data to identify concepts that require additional practice or support (Strielkowski et al., 2024).

By monitoring students' progress throughout the learning process, these systems can provide timely feedback and recommend appropriate learning activities based on individual needs. This enables students to recognize their learning gaps, focus on areas that need improvement, and develop a stronger understanding of concepts while studying independently.

The reviewed literature highlights the important role of Artificial Intelligence in enhancing education by providing personalized learning experiences, adapting instructional content based on students' performance, and identifying learning gaps that require additional support. These capabilities address the limitations of traditional learning approaches, which often provide the same learning experience to all students regardless of their individual needs and progress.

Guided by these findings, LearnSync AI integrates Artificial Intelligence to deliver adaptive and personalized academic support for Grade 10 students. By continuously adjusting learning activities according

to students' progress and identifying areas that require reinforcement, the platform promotes more effective independent learning and encourages deeper conceptual understanding. Through this adaptive approach, LearnSync AI aims to provide a responsive learning environment that improves student engagement, strengthens knowledge retention, and supports better learning progress.

Combining Gamification and Evidence-Based Learning Strategies

Modern digital learning platforms increasingly combine multiple instructional approaches to support different aspects of student learning. Gamification enhances motivation and engagement by incorporating game elements that encourage active participation and sustained learning (Sailer & Homner, 2020).

At the same time, evidence-based cognitive learning strategies, particularly retrieval practice, strengthen understanding, memory, and long-term knowledge retention by encouraging learners to actively recall previously learned information (Agarwal et al., 2021). Rather than serving the same purpose, these approaches complement one another by addressing different learning needs.

Recent systematic reviews suggest that well-designed gamified learning environments are most effective when game elements are integrated with sound instructional strategies that promote meaningful learning rather than functioning as isolated activities (Dehghanzadeh et al., 2024). These findings support the integration of gamification and evidence-based learning strategies within a single digital learning platform to provide engaging and meaningful learning experiences.

Gamification Procedures

Gamification has become an effective teaching approach that improves student engagement, motivation, and learning by incorporating game elements into educational activities. Unlike traditional study methods that rely mainly on reading or memorization, gamified learning encourages students to participate actively through interactive challenges, immediate feedback, and achievable goals. Studies have shown that gamification can enhance students' motivation, engagement, and learning experiences (Sailer & Homner, 2020).

In digital learning platforms, gamified learning activities provide learners with interactive opportunities to

review previously learned concepts while maintaining motivation during independent study. By integrating game elements with educational objectives, these activities encourage active participation and sustained engagement rather than functioning only as entertainment. Systematic reviews have shown that well-designed gamification supports meaningful learning experiences by increasing learners' motivation and participation when aligned with sound instructional strategies (Sailer & Homner, 2020; Dehghanzadeh et al., 2024).

Shared Mechanism

Crossword puzzles, matching cards, and timed trivia questions are connected because they all use the same basic learning process: students must remember and retrieve information, not just reread it, and gamification research shows that this kind of activity can improve learning, motivation, and participation (Monzon & Hays, 2024; Dahalan et al., 2023). They also share common game features such as challenge, immediate feedback, and clear goals, which help keep students engaged during review activities (Al-Khresheh, 2025). Crossword studies show that

puzzles improve terminology recall, vocabulary retention, and active participation because students solve clues by remembering meanings and spellings (Jamal et al., 2026; Ramadhani & Shofiyuddin, 2025; Hsu et al., 2023). Matching card activities show a similar pattern because students must connect related ideas from memory, and card-based learning has been found to improve retention, interest, and confidence (Yu et al., 2025; Gursesli et al., 2023).

Timed trivia is related to the other two because it also depends on fast recall, repeated practice, and immediate correction of answers, which are features linked to better memory and attention in gamified learning (Sailer & Sailer, 2020; Al-Khresheh, 2025). In simple terms, the three games are different in format, but they do the same job: crossword strengthens word and term recall, matching cards strengthen associations between related concepts, and timed trivia strengthens quick retrieval and feedback-based review (Jamal et al., 2026; Yu et al., 2025; Hsu et al., 2023).

Crossword

Crossword puzzles have become a useful educational activity for reinforcing knowledge and

improving students' understanding of key concepts. Unlike traditional review methods that rely only on reading or memorization, crossword puzzles encourage learners to actively recall information by identifying and connecting terms with their corresponding definitions or clues. This process promotes deeper cognitive engagement while making learning more interactive and enjoyable.

Research suggests that crossword puzzles can improve students' understanding of key concepts and support better performance in objective assessments by encouraging active recall during learning activities (Arnold et al., 2024). These findings suggest that incorporating crossword puzzles into digital learning platforms can strengthen concept retention while increasing student engagement during independent study.

Matching Card Games

Matching card games are widely used as interactive learning activities that promote active participation and reinforce students' understanding of

concepts. By requiring learners to match related terms, definitions, images, or concepts, these activities encourage active recall and strengthen associations between pieces of information rather than relying on passive reading.

Recent research found that matching card-based learning materials significantly improved students' vocabulary mastery and learning outcomes by providing an engaging and interactive learning experience (Febrianti & Shofiyuddin, 2025). Because matching activities encourage learners to repeatedly recall and associate related concepts, their underlying principles can be adapted to reinforce foundational knowledge in digital learning environments.

Timed Trivia Questions

Timed trivia questions are an effective gamified learning activity that combines retrieval practice with interactive quiz-based learning. By requiring learners to answer review questions within a limited time, students actively retrieve information from memory while maintaining focus under structured time

constraints. This transforms conventional review sessions into engaging learning experiences that encourage continuous participation and immediate feedback. A literature review found that game-based quiz platforms such as Kahoot! consistently improve student engagement, motivation, classroom participation, and learning performance across different educational settings (Wang & Tahir, 2020).

In addition, the integration of game elements such as time limits, scores, and immediate feedback has been shown to positively influence learners' motivation and academic outcomes when appropriately incorporated into educational activities (Sailer & Homner, 2020). Together, these findings suggest that timed trivia questions provide an engaging review strategy that encourages active participation while reinforcing previously learned concepts through repeated retrieval practice.

Retrieval Practices

Cognitive retrieval practices are an important component of modern educational psychology that enhance

how students recall and retain learned information. Unlike traditional study methods that rely on rereading or passive memorization, retrieval-based learning requires students to actively recall information from memory, which strengthens long-term retention and improves understanding of concepts. Research has consistently shown that retrieval practice is more effective than passive review strategies in improving learning and long-term retention (Agarwal et al., 2021).

In addition, testing-based learning has been found to significantly enhance classroom learning by improving memory retention and reinforcing conceptual understanding over time (Yang et al., 2021). Retrieval practice can be implemented through learning strategies such as free recall and structured explanation exercises, including the Feynman Technique, which encourage learners to actively reconstruct knowledge from memory.

Feynman Technique

The Feynman Technique is a learning strategy that helps students develop a deeper understanding of concepts by explaining lessons in their own words using simple and clear language. Rather than relying on memorization, learners actively recall information,

identify gaps in their understanding, and revisit concepts until they can explain them accurately. This process encourages students to organize and connect ideas, promoting meaningful learning instead of rote memorization.

Research on generative learning suggests that explanation-based learning helps students construct knowledge by organizing information, connecting new ideas with prior knowledge, and actively generating explanations. These learning processes strengthen conceptual understanding, support long-term retention, and encourage learners to monitor their own understanding throughout the learning process (Fiorella, 2023). Likewise, Reyes et al. (2021) found that students who practiced explaining concepts in simple language demonstrated better conceptual understanding than those who used traditional study methods. These findings suggest that the Feynman Technique is an effective cognitive learning strategy for supporting comprehension, identifying knowledge gaps, and promoting independent learning.

Free Recall

Free recall is a form of retrieval practice that helps students strengthen their memory by encouraging them to remember and explain what they have learned without using notes or other learning materials. Instead of simply rereading information, students actively retrieve knowledge from memory, making it easier to identify topics they understand and concepts that require further review. This process improves understanding and supports long-term retention (Yang et al., 2021).

Research has shown that retrieval practice, including free recall, is more effective than passive study methods such as rereading for improving learning and memory (Agarwal et al., 2021; Yang et al., 2021). Actively recalling information improves memory, strengthens understanding, and supports deeper understanding and long-term retention. (Agarwal et al., 2021). Likewise, retrieval-based learning helps students retain information longer and reinforces their understanding of lesson concepts (Yang et al., 2021). These findings suggest that free recall is an effective cognitive learning strategy for supporting students' independent learning.

Adaptive Flashcards

Flashcards are widely used as a learning tool that supports active recall and improves memory retention. Unlike passive study methods such as rereading notes, flashcards require learners to actively retrieve information from memory, which strengthens understanding and reinforces learned concepts. Research on vocabulary learning has shown that intentional use of flashcards can significantly improve recall performance and learning outcomes compared to traditional study methods such as word lists and repeated reading (Webb et al., 2020).

In addition, studies on digital and mobile-assisted learning indicate that flashcards are effective in supporting vocabulary acquisition and improving students' retention of academic words when used as part of structured study activities (Xodabande et al., 2022). The effectiveness of flashcards can be further enhanced through spaced repetition, a learning technique that schedules review sessions based on learners' previous performance and recall. By increasing review intervals for well-learned concepts while providing more frequent review for difficult

concepts, spaced repetition improves long-term retention and learning efficiency (Settles & Meeder, 2020).

Modern adaptive learning systems further improve this process by using data-driven approaches to personalize review schedules. Unlike static study methods where learners may repeatedly review familiar material or neglect difficult concepts, adaptive flashcard systems adjust the frequency of review based on student performance. These systems help optimize learning efficiency by focusing more on challenging items and reducing unnecessary repetition, which can improve long-term retention and reduce cognitive load (Huang, 2025).

Adaptive Assessment

Computerized adaptive testing has become an important development in modern educational assessment, shifting traditional fixed testing formats toward more flexible and data-driven evaluation systems. Unlike conventional assessments where all students answer the same set of questions regardless of ability, adaptive testing systems adjust question difficulty based on learners' responses and performance patterns. Recent research highlights that these

systems use real-time performance data and machine learning techniques to select appropriate test items, allowing assessments to better match each student's level of understanding (Zhuang et al., 2024).

In addition, AI-driven quiz generation systems have been shown to support the creation of dynamic assessments that can adjust question difficulty and provide more personalized evaluation experiences (Al-Gburi, 2025). Through these adaptive mechanisms, computerized testing systems aim to improve the efficiency of assessments and provide more accurate representations of student knowledge and learning progress.

The reviewed literature suggests that integrating gamification, retrieval-based learning strategies, adaptive flashcards, and adaptive assessment can contribute to more effective and engaging learning experiences compared to traditional passive study methods. Gamification has been shown to enhance student motivation and engagement by incorporating interactive and goal-oriented learning activities, while cognitive strategies such as active recall and retrieval practice support deeper understanding and long-term retention of information. Likewise, spaced repetition and adaptive flashcards promote more efficient review

by prioritizing concepts that require additional practice while reducing unnecessary repetition of concepts that learners have already mastered. In addition, adaptive assessment approaches improve learning efficiency by adjusting the difficulty and content of evaluation tasks based on student performance, allowing learners to be assessed according to their current level of understanding.

Collectively, these approaches highlight the value of combining motivational, cognitive, and adaptive learning strategies within digital learning environments. Guided by these findings, LearnSync AI integrates spaced repetition, Bayesian Knowledge Tracing–based adaptive flashcards, adaptive assessments, retrieval-based learning strategies, and gamified learning activities into a unified platform that supports personalized, adaptive, and self-directed learning. Through this integrated approach, the platform aims to promote student engagement, reinforce knowledge retention, and support independent learning.

Performance Evaluation of Language Models and Inference Systems

The performance of large language models and their inference systems is an important factor in ensuring that AI-powered educational platforms provide fast, reliable, and responsive learning experiences. In digital learning environments,

delays in processing user inputs or generating responses can negatively affect student engagement and disrupt learning continuity. As a result, modern LLM systems are designed with optimized inference and serving architectures that improve how models handle multiple user requests efficiently. Research on distributed model serving systems shows that optimized inference techniques can improve system throughput and response efficiency in transformer-based language models (Yu et al., 2022).

In addition, studies on memory-efficient LLM serving highlight that improved resource management and computational optimization help reduce latency and support smoother real-time interactions in large-scale AI applications (Kwon et al., 2023). These advancements are particularly important for educational platforms that rely on generative AI, as they ensure timely feedback, stable performance, and uninterrupted learning support for users.

LLM in Materials and Context

The use of large language models (LLMs) in educational systems plays an important role in improving how students interact with complex and lengthy learning materials. Early transformer-based language models

demonstrated strong capabilities in understanding and generating human-like text, enabling flexible language processing across a wide range of inputs (Brown et al., 2020).

While earlier transformer-based language models were limited by relatively restrictive context windows that affected continuity in multi-topic interactions, recent advancements in large language models have significantly improved their ability to process longer contextual inputs. These improvements allow models to generate more coherent and context-aware responses when working with extended learning materials. As a result, when integrated into educational platforms, LLM-based systems can better support student understanding of complex topics by providing more consistent and contextually relevant explanations during independent study (Sharma et al., 2025).

Real Time Feedback

Large language model–based systems play an important role in providing immediate formative feedback within digital learning environments. In traditional systems, delays in generating responses may interrupt students' focus

during active learning tasks such as retrieval practice and review activities. In contrast, modern AI-powered systems are capable of delivering near real-time feedback and personalized guidance, which supports student engagement and sustained participation in learning activities (Kasneci et al., 2023).

Reducing response delays also helps create a smoother interaction between the learner and the system, minimizing disruption during study sessions. This immediate feedback allows students to quickly identify and correct mistakes, which can improve learning efficiency, task engagement, and revision behavior (Meyer et al., 2024).

Model Specialization and Dynamic Routing

The use of multiple large language models within a single system plays an important role in improving the efficiency and responsiveness of AI-powered applications. Instead of relying on a single model to handle all types of tasks, modern systems can distribute requests by assigning them to different models based on task complexity and computational requirements. Research on LLM routing systems shows that queries can be directed to different

models depending on expected performance needs, allowing systems to balance quality and efficiency (Ong et al., 2024).

In this approach, simpler or less computationally demanding tasks can be handled by more efficient models, while more complex requests are processed by higher-capacity models. This routing strategy helps optimize system performance by reducing unnecessary computational load and improving response time, resulting in a smoother and more efficient user experience in real-time applications.

The reviewed studies demonstrate that an effective AI-powered educational platform depends not only on the capabilities of large language models but also on efficient inference systems, real-time feedback, contextual understanding, and appropriate task distribution among AI components. Together, these technologies create a learning environment that is responsive, reliable, and capable of supporting students throughout their independent study sessions.

Guided by these findings, LearnSync AI integrates these concepts to provide fast AI-generated assistance, process lengthy study materials effectively, deliver immediate personalized feedback, and maintain smooth system performance. By combining this into a unified application, it provides Grade 10 students with a more efficient, engaging, and

personalized learning experience that supports independent study and improves overall learning outcomes.

Bayesian Knowledge Tracing

Bayesian Knowledge Tracing (BKT) is a probabilistic model used in intelligent tutoring systems to estimate a learner's mastery of specific skills based on their sequence of correct and incorrect responses. It works by continuously updating the probability that a student has learned a concept, making it useful for modeling knowledge progression in adaptive learning environments.

BKT is commonly used in educational systems due to its simplicity, interpretability, and ability to support dynamic learner modeling without requiring high computational complexity (Bulut et al., 2023).

Tracking Models: BKT, DKT, and PFA

In educational data mining, predicting student learning progression involves modeling hidden knowledge states using different algorithmic approaches, including Bayesian Knowledge Tracing (BKT), Performance Factors Analysis

(PFA), and Deep Knowledge Tracing (DKT) (Abdelrahman et al., 2023). These models vary in structure and complexity. BKT is a probabilistic model that estimates student mastery based on sequential performance, while PFA is a statistical approach that predicts learning outcomes by considering the cumulative count of prior successes and failures in skill practice. In contrast, DKT utilizes deep learning techniques to model complex learning patterns from large-scale datasets.

Research indicates that deep learning-based approaches such as DKT can achieve strong predictive performance, particularly when large datasets are available, but they may introduce challenges related to interpretability compared to traditional models (Scruggs et al., 2020). Meanwhile, BKT and PFA remain widely used in educational systems due to their simplicity, interpretability, and lower computational requirements, making them suitable for applications that require transparent learner modeling. These differences highlight the trade-offs between predictive accuracy, interpretability, and computational complexity when selecting knowledge tracing models for educational applications

BKT Selection

Among the different knowledge tracing models reviewed, including Performance Factors Analysis (PFA) and Deep Knowledge Tracing (DKT), the researchers selected Bayesian Knowledge Tracing (BKT) as the learner modeling approach for the system. This decision is based on its suitability for adaptive learning environments that require efficient, interpretable, and sequential tracking of student knowledge.

Unlike DKT, which requires large datasets and higher computational resources, and PFA, which focuses on aggregated performance counts rather than step-by-step learning progression, BKT provides a lightweight framework that updates learner mastery based on individual responses. Its simplicity and transparency make it appropriate for educational applications where clear interpretation of student progress is important. Therefore, BKT was chosen to support the system's need for continuous, real-time, and interpretable tracking of learner performance in adaptive learning activities such as flashcards and assessments.

The integration of Bayesian Knowledge Tracing (BKT) into LearnSync AI provides an efficient method for tracking student mastery during independent study sessions. By continuously updating a learner's knowledge state based on their responses, the system can estimate skill mastery while accounting for random factors such as guessing and slipping. Its lightweight learner modeling approach also helps maintain fast and responsive system performance.

Furthermore, BKT enables LearnSync AI to personalize adaptive learning activities by adjusting the difficulty of the Adaptive Flashcards and Adaptive Test Mode according to the learner's estimated mastery level. This supports a more individualized independent learning experience while maintaining an interpretable and computationally efficient learner model.

SIGNIFICANCE OF THE STUDY

This study proposes LearnSync AI, a personalized learning platform designed to enhance independent learning through personalized learning support, adaptive learning activities, AI-assisted study strategies, gamified learning experiences, and learning analytics. The proposed system is expected to benefit the following:

Pasig Christian Academy. The proposed system will be a major advantage for Pasig Christian Academy by providing a centralized, AI-assisted learning platform that complements the school's existing independent learning process. LearnSync AI enables the institution to integrate adaptive learning, AI-assisted learning support, personalized assessments, and verified instructional materials into a secure learning environment. Because the platform is designed strictly for asynchronous, independent study at home, it acts as a modern extension of the curriculum without disrupting the school's traditional, face-to-face daily operations.

Educators. The proposed system provides educators with a secure and centralized platform for uploading and managing official learning resources used by LearnSync AI. By ensuring that AI-generated learning support is based only on teacher-approved instructional materials, the system helps maintain curriculum alignment and academic accuracy. In addition, the educator dashboard allows teachers to view students' learning progress and mastery statistics, enabling them to gain insights into students' independent learning without significantly increasing their workload.

Junior High School Students. The proposed system provides Junior High School students with a personalized study companion that customizes learning activities based on their current skill levels. By combining adaptive learning activities, AI-assisted feedback, gamified learning experiences, and personalized learning analytics in a single platform, the LearnSync AI encourages students to become more engaged in independent learning while

helping them monitor their mastery progression and strengthen their study habits. This ensures students truly understand their key lessons and build strong study habits as they prepare for the academic demands of Senior High School.

Researchers. The study will contribute to the development of the researchers' knowledge and skills in the fields of software development, educational technology, artificial intelligence, adaptive learning systems and Bayesian Knowledge Tracing. It will also give them good practice in designing, developing, implementing and evaluating an AI-based educational platform.

Future Researchers. The findings of this study may serve as a reference for future researchers developing AI-powered educational platforms. It may also provide a foundation for further studies on adaptive learning, Generative Artificial Intelligence, Bayesian Knowledge Tracing, and personalized learning technologies across different educational contexts.

STATEMENT OF THE PROBLEM

As Junior High School students progress through increasingly complex academic subjects, independent learning becomes essential in reinforcing classroom instruction and preparing them for the academic demands of Senior High School. At Pasig Christian Academy, students primarily conduct independent study using educator-provided learning materials and official textbooks. While these resources effectively support the school's curriculum, the current independent learning process provides limited opportunities for

personalized learning, adaptive learning support, and continuous monitoring of students' learning progress. To complement the school's existing independent learning process, the researchers propose LearnSync AI, an educational platform that integrates Generative Artificial Intelligence (Generative AI) and Bayesian Knowledge Tracing (BKT) into a single platform that provides personalized learning support, adaptive learning experiences, and continuous monitoring of students' learning progress.

This study seeks to evaluate the platform's technical accuracy, support for independent learning, user effectiveness, and overall acceptability through the following research questions:

1. What challenges do Junior High School students encounter during independent learning at Pasig Christian Academy?

2. How accurately does LearnSync AI implement Generative Artificial Intelligence (GenAI) and Bayesian Knowledge Tracing (BKT) in generating personalized learning content and adaptive learning activities?

3. What is the measurable improvement in the knowledge mastery of Junior High School students after utilizing LearnSync AI's personalized study features, as reflected by their Bayesian Knowledge Tracing (BKT)

progression?

4. What is the level of acceptability of LearnSync AI as evaluated by the intended users, specifically Junior High School students and educators based on the functionalities accessible to them, in terms of the following eight ISO/IEC 25010 software quality characteristics:

    - Functional Suitability
    - Performance Efficiency
    - Compatibility
    - Usability
    - Reliability
    - Security
    - Maintability
    - Portability

SCOPE AND LIMITATIONS

This study focuses on the design, development, implementation, and evaluation of LearnSync AI, an AI-powered educational platform that supports the independent learning of Junior High School students through adaptive learning, Generative Artificial Intelligence (Generative AI), Bayesian Knowledge Tracing (BKT), gamification, and learning analytics in a single platform.

Scope of the Study:

This study is limited to the implementation and evaluation of LearnSync AI involving 50 selected Junior High School students (Grades 7 - 10) of Pasig Christian Academy during the Academic Year 2026 - 2027. Although the respondents come from different grade levels, they will be analyzed as a single study population. Comparative analyses among grade levels are outside the scope of the study. Instead, the evaluation focuses on each student's individual learning progress and knowledge mastery throughout the implementation period.

The platform is designed to complement the school's existing learning process by supporting students' independent study outside regular classroom instruction. It is not intended to replace traditional classroom teaching or disrupt scheduled learning activities. LearnSync AI serves as an extension of the school's curriculum by providing AI-assisted learning experiences using educator-approved instructional materials.

The platform includes an educator module that enables authenticated educators to upload and manage official learning resources and view student learning statistics. These uploaded

materials serve as the exclusive knowledge base for all AI-assisted functionalities, ensuring that generated content remains aligned with the school's curriculum while maintaining academic accuracy and consistency.

LearnSync AI integrates the classical four-parameter Bayesian Knowledge Tracing (BKT) model to estimate each learner's mastery level based on their performance in Adaptive Flashcards and Adaptive Test Mode. These mastery estimates are then used to personalize learning activities and dynamically adjust the question selection and difficulty of AI-generated educational games, including Crossword, Term Matching, and Timed Trivia.

The platform further supports independent learning through Generative Artificial Intelligence (GenAI) by providing AI-assisted study strategies such as the Feynman Technique and Free Recall, personalized feedback, and an AI-powered learning assistant capable of answering questions based exclusively on educator-uploaded learning resources.

Finally, the study evaluates LearnSync AI using the applicable ISO/IEC 25010 Software Quality Model, specifically the characteristics of Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

Limitations of the Study:

The implementation and evaluation of LearnSync AI are limited to a 2 to 4 week period involving 50 selected Junior High School students of Pasig Christian Academy. Because of this limited implementation period, the study evaluates only short-term knowledge mastery progression during system use and does not measure long-term knowledge retention, sustained learning behaviors, long-term academic achievement, or permanent changes in student performance.

To ensure a manageable implementation, the learning activities will initially focus on the subject identified through the requirement analysis as the most difficult for students in each participating grade level. Although respondents come from different grade levels, the study does not perform comparative analyses among Grades 7, 8, 9, and 10. Instead, all participants are analyzed as a single study population, with emphasis placed on each student's individual mastery progression throughout the implementation period.

The platform employs only the classical Bayesian Knowledge Tracing (BKT) model. Student mastery estimates are updated exclusively through their performance in Adaptive

Flashcards and Adaptive Test Mode. AI-generated educational games utilize existing mastery estimates only to personalize question selection, generated content, and difficulty level without modifying the learner's mastery state. Likewise, the AI-assisted Feynman Technique and Free Recall provide personalized explanations and feedback but do not directly update BKT mastery estimates.

The platform experiences an inherent algorithmic cold-start limitation whenever a new learner begins using the system or a new learning module is introduced. Since the Bayesian Knowledge Tracing (BKT) model initially has no prior evidence of a learner's knowledge state, all students begin with the same default mastery probabilities across all learning concepts. Consequently, personalization is not immediate and becomes progressively more representative of each learner's actual knowledge as they complete Adaptive Flashcards and Adaptive Test Mode throughout the implementation period.

The AI-powered learning assistant generates responses exclusively from educator-uploaded learning resources. Consequently, the quality, completeness, and accuracy of AI-generated explanations depend entirely on the uploaded instructional materials. The chatbot cannot generate information

beyond the available learning resources or external knowledge sources.

Operationally, LearnSync AI requires a stable internet connection for all online functionalities. The platform accepts plain-text PDF learning modules with a maximum file size of 20 MB. To accommodate the school's continued reliance on printed textbooks and other physical instructional materials, the platform also supports image-based or scanned PDF documents with a maximum of 20 pages, which are processed for use as AI learning resources. To preserve academic integrity and ensure curriculum alignment, only authenticated educators are authorized to upload learning resources. Students are not permitted to upload instructional materials, and all AI-assisted functionalities depend entirely on the availability of educator-provided learning resources. The platform does not support offline functionality, handwritten text recognition, voice-based interaction, payment processing, automated billing, or enterprise-level server scalability.

CHAPTER II

INTRODUCTION

This chapter presents the methodology used to design, develop, and evaluate LearnSync AI: AI Personalized Learning Companion for Junior High School Students. This web-based platform is created for Junior High School (JHS) students to support them in their independent learning, with an accompanying Educator Portal that gives teachers direct control over the learning materials their students access. The sections in this chapter detail the system's technical design, development workflows, algorithms, and data structures that allow the platform to operate safely, securely, and efficiently.

The application is designed with two distinct user roles. The Educator uploads and manages the learning materials that students will use, ensuring that all study content has been reviewed and validated by a qualified teacher before it reaches the learning environment. The Educator can also monitor the learning mastery progress of their students through a class-level analytics dashboard. The Student accesses these educator-provided materials and engages with the platform's AI-assisted learning features, including Adaptive Flashcards, Adaptive Test Mode, the Feynman Technique, Free Recall, AI-generated educational games, and a personal analytics dashboard. Bayesian Knowledge Tracing (BKT) continuously estimates each student's mastery and uses the resulting scores to personalize the difficulty of learning activities.

Finally, this chapter describes the system implementation and evaluation procedures. A 2 to 4 week implementation and user evaluation will be conducted at Pasig Christian Academy with Junior High School students and their assigned educators serving as the study participants. User feedback will be collected using an evaluation instrument based on the ISO/IEC 25010 software quality model to assess the platform's Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. The results will be used to determine the system's quality and overall acceptability as an AI-assisted learning platform.

RESEARCH DESIGN

This study uses a developmental research design with a quantitative approach to guide the design, construction, and evaluation of the LearnSync AI web application. In educational technology, developmental research focuses on the systematic creation, testing, and evaluation of instructional tools to ensure they are stable, functional, and effective (Angeles & Cortez, 2023). By using this design, this project transitions from theoretical software planning to a functional, real-world application tested by its target users. The systematic recording of the design and development processes allows the research team to translate educational needs into functional software features while collecting objective numbers to prove the application's overall performance.

The developmental process for LearnSync AI is structured into three distinct phases. The first phase, analysis and design, identifies the study challenges faced by Junior High School students during self-paced study and the supervisory needs of their educators, then uses these findings to design the user interface layouts of the Student Portal and Educator Portal, map out the study space architecture, establish the BKT routing rules, and draft the PlayMe gamification mechanics. The second phase, development, involves the technical implementation of the web application, including configuring the Supabase relational database with Row-Level Security policies, programming the generative AI parser to extract study terms from educator-uploaded materials, implementing the real-time BKT scoring engine, and building the educator-facing class analytics dashboard. The third phase, evaluation, deploys the finished platform over a 2 to 4 week period at Pasig Christian Academy, where educators will upload learning materials through the Educator Portal while Junior High School students independently use the platform's learning features. Participant experiences and feedback will be collected through a structured evaluation questionnaire to assess quality and user acceptability.

To objectively evaluate the developed system, the study utilizes quantitative data collected through two complementary methods. The first method involves system-generated operational data, including students' interactions with Adaptive Flashcards and Adaptive Test Mode, where the Bayesian Knowledge Tracing (BKT) algorithm continuously estimates learner mastery based on assessment performance. These data are used to validate the

operation of the adaptive learning mechanism and verify that the system personalizes learning activities according to students' estimated mastery.

The second method is a software quality evaluation based on the ISO/IEC 25010 software quality model. After using the platform, the participants will complete a structured evaluation questionnaire using a four-point Likert scale. The collected responses will be analyzed using descriptive statistics to determine the level of acceptability of LearnSync AI in terms of Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. The results will be used to assess the overall quality and user acceptability of the developed platform.

ARCHITECTURAL FRAMEWORK

The architectural framework of LearnSync AI follows a dual-portal, three-tier model. The Educator Portal handles material upload, grade-level access control, and mastery monitoring. The Student Portal handles independent study, adaptive assessment, and gamified review. The system is divided into three tiers: Presentation, Application, and Data, with information flowing down when a user performs an action and back up when the system returns a result.

1. **Presentation Tier -** This is the interface layer that both educators and students interact with through their web browser. It has four parts. The

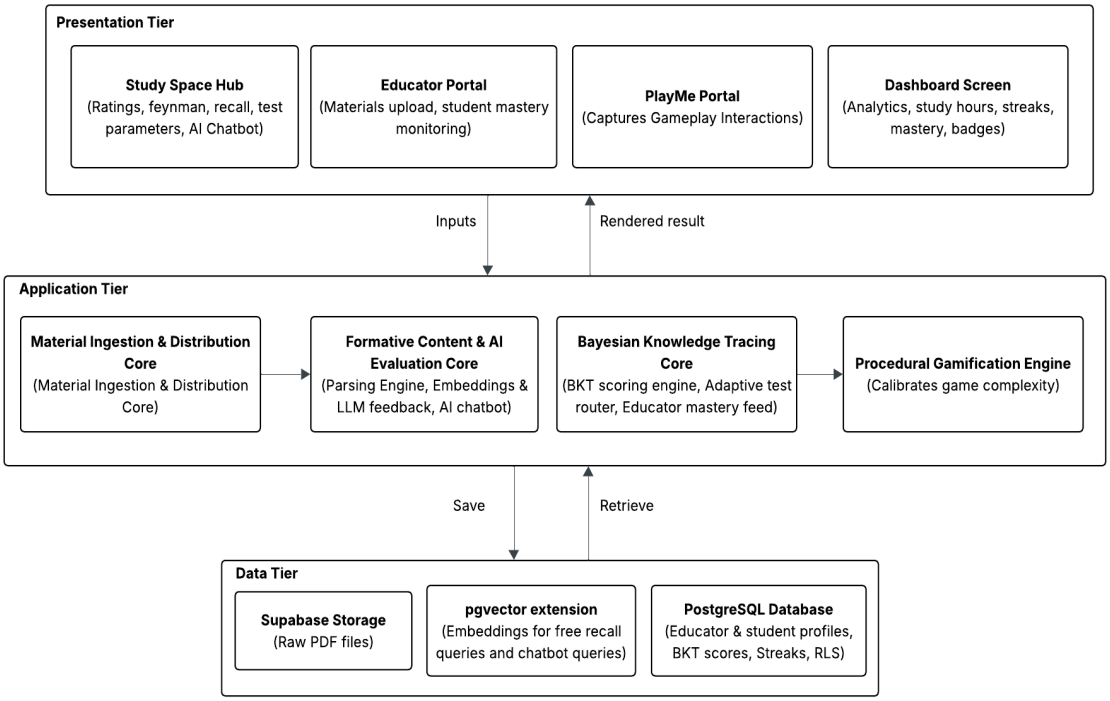
Educator Dashboard allows the educator to upload PDF learning materials, select which grade level can access them, and view a real-time analytics overview of individual student mastery levels. The Study Space Hub is where students access educator-uploaded materials, rate flashcards, complete Feynman Technique and Free Recall exercises, configure and take practice tests, and interact with the AI chatbot. The PlayMe Portal captures student interactions during Crossword, Term Matching, and Timed Trivia games. The Dashboard Screen displays the student's personal progress, including study hours, daily streaks, mastery curves, and earned badges.

2. **Application Tier -** This is the processing layer. It has four parts. The Material Ingestion and Distribution Core receives educator-uploaded documents, routes them through the parsing engine, and makes the processed content available to students within the educator's selected grade level. The Formative Content and AI Evaluation Core extracts key terms and concepts from parsed materials, evaluates student Free Recall and Feynman responses using vector embeddings and LLM feedback, and powers the AI chatbot responses grounded in the uploaded learning resource. The Bayesian Knowledge Tracing (BKT) Core takes flashcard ratings and test answers, calculates per-concept mastery estimates, routes adaptive test difficulty through the Adaptive Test Router, and feeds aggregated mastery data back to the Educator & Student Dashboard. The

Procedural Gamification Engine reads the BKT mastery score to calibrate game difficulty to the student's current level.

3. **Data Tier -** This is the storage layer, running on Supabase. Supabase Storage holds the raw PDF files uploaded by educators. The pgvector extension stores vector embeddings used for Free Recall similarity checks and AI chatbot retrieval. The PostgreSQL Database holds educator and student profiles, grade-level material access rules, BKT mastery scores, and study streaks, with Row-Level Security (RLS) enforcing data isolation between students and restricting educators to the mastery data of students within their selected grade level.

Figure 1. *Architectural Framework*



ALGORITHM

This section details the mathematical and algorithmic foundation of LearnSync AI's cognitive modeling core. To deliver a highly personalized, self-directed learning experience, the platform integrates the classical four-parameter Bayesian Knowledge Tracing (BKT) model. BKT is a probabilistic learner model that has remained a widely used and interpretable approach for tracing student knowledge, and it continues to be validated in modern educational technology research, including through accessible implementations such as the pyBKT framework (Bulut et al., 2023). Within LearnSync AI, the four-parameter update equations are implemented directly in the platform backend. BKT operates as a dynamic hidden Markov model (HMM) with observable nodes, continuously analyzing a student's sequential interactions to trace their latent knowledge state for any given concept without requiring computationally heavy server resources (Abdelrahman et al., 2023).

1. BKT Mathematical Model - The primary objective of the BKT algorithm is to estimate the probability that a student has mastered a specific concept based on their chronological sequence of correct and incorrect answers. The mathematical framework is governed by four distinct, highly interpretable parameters categorized into two knowledge parameters and two performance parameters (Bulut et al., 2023):

a. **Initial Knowledge State** $P(L_0)$: The prior probability that a student already knows or has mastered a concept before interacting with any formative study modes or assessments on the platform.
b. **Transition Probability** $P(T)$: The learning rate represents the probability that a student shifts from an unmastered (not-known) state to a mastered (known) state after a learning opportunity.
c. **Guess Probability** $P(G)$: The probability that the student has not mastered the concept but is still able to answer correctly.
d. **Slip Probability** $P(S)$: The probability that the student has mastered the concept still answers incorrectly due to carelessness, cognitive fatigue, or a reading error.

As the student interacts with the application, their performance at any given learning opportunity *t* is captured as an observable binary variable (obs_t), where an answer is marked as correct $(obs_t = 1)$ or incorrect $(obs_t = 0)$. When a new observation is captured, the system executes a two-phase Bayesian

update. First, the algorithm applies the evidence update phase to calculate the conditional probability of mastery ($P(L_t \mid obs_t)$) based on the observation:

Formula when the students attempt is correct ($obs_t = 1$) :

$$\textit{EQ (1):}\ P(L_t \mid obs_t = 1) = \frac{P(L_t)\,(1-P(S))}{P(L_t)\,(1-P(S)) + (1-P(L_t))\,P(G)}$$

Formula when the students attempt is incorrect ($obs_t = 0$) :

$$\textit{EQ (2):}\ P(L_t \mid obs_t = 0) = \frac{P(L_t)\,P(S)}{P(L_t)\,P(S) + (1-P(L_t))\,(1-P(G))}$$

Second, the algorithm performs the transition update phase. This step calculates the projected probability of mastery for the subsequent learning opportunity $P(L_{t+1})$ by factoring in the learning rate $P(T)$ as the student moves to the next card or question, assuming immediate formative feedback is processed:

$$\textit{EQ (3):}\ P(L_{t+1}) = P(L_t \mid obs_t) + (1 - P(L_t \mid obs_t))\,P(T)$$

1.1. Default Parameter Initialization - Since LearnSync AI is a newly developed platform with no pre-existing historical student interaction data, the system cannot utilize data-driven parameter estimation methods such as Expectation Maximization (EM) fitting, which requires large datasets of prior student responses to automatically learn optimal parameter values (Bulut et al., 2023). Instead, the system initializes BKT parameters using empirically grounded default values derived from established educational data mining research. These values serve as reasonable starting points that allow the BKT model to begin tracking mastery immediately upon a student's first interaction, and the model's Bayesian updating mechanism ensures that mastery estimates self-correct and converge toward accurate values as more interaction data accumulates over time (Bulut et al., 2023).

The following default parameter values are adopted for all newly created concept nodes upon document upload:

a. $P(L_0) = 0.10$. The initial knowledge state is set to 0.10, reflecting the conservative assumption that a student has a 10% probability of having already mastered any given concept before engaging with the platform. This low prior is consistent with the standard low-prior initialization used for educational knowledge tracing applications, ensuring that the system does not overestimate a student's starting competency and allows mastery to grow organically through demonstrated performance (Bulut et al., 2023).

b. $P(T) = 0.20$. The transition probability (learning rate) is set to 0.20, indicating a 20% probability that a student transitions from an unmastered state to a mastered state after each learning opportunity. Because LearnSync AI operates as a self-directed, independent learning platform, a moderately conservative learning rate is adopted to reflect the additional self-monitoring burden that learners carry when instructor feedback is absent (Luo & Zhou, 2024). As with all defaults, the Bayesian updating mechanism refines this estimate as evidence accumulates (Bulut et al., 2023).

c. $P(G)$ = Context-Sensitive (Dynamic). Unlike the standard BKT model, where the guess probability is a fixed constant per knowledge component, LearnSync AI introduces context-sensitive guess parameters that vary based on the assessment mode, question format, and the student's self-reported confidence level. This adaptation is mathematically valid because the BKT evidence-update formulas accept any valid P(G) value between 0 and 1 on a per-observation basis without violating the model's probabilistic assumptions (Bulut et al., 2023).

In Adaptive Flashcard Mode, the guess probability varies by the student's self-reported confidence rating. A rating of Easy assigns $P(G_easy) = 0.05$, reflecting high confidence that the correct response was due to genuine knowledge rather than guessing. A rating of Medium assigns $P(G_medium) = 0.25$, accounting for a moderate possibility that the response was aided by partial guessing. A rating of Hard assigns $P(G_hard) = 0.10$,

acknowledging that if a student who forgot the card had somehow still produced a correct response, it would be attributable almost entirely to chance; this value is slightly above the 0.05 floor to allow for the possibility of vague partial recall, and it is used when the BKT incorrect-answer formula evaluates the $1 - P(G)$ term in the denominator (Bulut et al., 2023).

In Adaptive Test Mode, the guess probability is calibrated per question type based on the mathematical probability of arriving at a correct response through random guessing alone. Because the baseline chance of a correct random response is determined directly by the number of available options, these values follow from the item format itself (Bulut et al., 2023). The following $P(G)$ values are adopted for each question format:

1. Multiple Choice (4 options): $P(G_mc) = 0.25$. With four answer options and one correct answer, the baseline probability of guessing correctly is 1/4 = 0.25.
2. True or False: $P(G_tf) = 0.50$. With only two possible responses, the baseline probability of guessing correctly is 1/2 = 0.50. This higher guess rate ensures that correct responses on True or False items contribute a substantially smaller mastery boost compared to other question types, reflecting the greater role of chance in binary-choice formats.
3. Identification: $P(G_id) = 0.05$. Open-ended identification items require the student to produce the exact term from memory without any provided options. The probability of guessing the correct term without prior knowledge is negligibly low.

4. Fill in the Blank: $P(G_fib) = 0.05$. Similar to identification items, fill-in-the-blank questions require free recall of specific terms or phrases without cues. The absence of selectable options makes random guessing highly unlikely, justifying $P(G) = 0.05$.
5. Matching Type (4 terms, 6 definitions): $P(G_match) = 0.17$. For any individual term, the probability of selecting the correct definition from the pool of six through random guessing is 1/6 ≈ 0.17, applied per matched item.

All adopted P(G) values remain below 0.50 and, for the flashcard and non-binary question types, below 0.30. Keeping guess and slip probabilities well below 0.50 preserves the interpretive validity of the model and avoids the degenerate parameter regions in which a correct answer would paradoxically lower the mastery estimate (Bulut et al., 2023). The True or False value necessarily approaches the 0.50 ceiling due to the inherent binary nature of the format, which the model accommodates on a per-observation basis.

d. $P(S) = 0.15$. The slip probability is set to 0.15, representing a 15% likelihood that a student who has genuinely mastered a concept may still answer incorrectly due to carelessness, misreading, or cognitive fatigue. This value moderately relaxes the classical $P(S) \leq 0.10$ constraint originally established for instructor-led tutoring systems (Bulut et al., 2023) to accommodate the self-directed learning context of the platform, where students are more susceptible to attentional lapses without real-time instructor monitoring (Luo & Zhou, 2024).

It is important to note that BKT is inherently self-correcting by design. Even if the initial default parameter values do not perfectly represent a given student's true learning behavior, the Bayesian updating mechanism continuously refines mastery estimates as more interaction data is collected. The default parameters therefore serve as empirically reasonable initialization points, and the system's accuracy improves organically as students accumulate more practice attempts across flashcard and test mode sessions (Bulut et al., 2023).

1.2. Step-by-Step Computation of Student Mastery - The following trace shows how the system updates a student's mastery of a single concept across three consecutive multiple-choice attempts using the default parameters: $P(L_0) = 0.10,\ \ P(T) = 0.20,\ \ P(S) = 0.15,\ \ P(G) = 0.25.$ Each attempt applies the evidence update first, then the transition update.

Attempt 1 - Correct $obs_t = 1$, prior $P(L_0) = 0.10$:

$$P(L_0 \mid obs_t = 1) = \frac{(0.10)(1-0.15)}{(0.10)(1-0.15)+(1-0.10)(0.25)} = 0.27$$

$$P(L_{0+1}) = 0.27 + (1 - 0.27)\,0.20 = 0.42$$

Attempt 2 - Correct $obs_t = 1$, prior $P(L_1) = 0.42$:

$$P(L_1 \mid obs_t = 1) = \frac{(0.42)(1-0.15)}{(0.42)(1-0.15)+(1-0.42)(0.25)} = 0.71$$

$$P(L_{1+1}) = 0.71 + (1 - 0.71)\,0.20 = 0.77$$

Attempt 3 - Incorrect $obs_t = 0$, prior $P(L_2) = 0.77$:

$$P(L_2 \mid obs_t = 0) = \frac{(0.77)(0.15)}{(0.77)(0.15)+(1-0.77)(1-0.25)} = 0.40$$

$$P(L_{1+1}) = 0.40 + (1 - 0.40)\,0.20 = 0.52$$

Two consecutive correct answers raised the student's mastery from 0.10 to 0.77, while a single subsequent incorrect answer reduced it to 0.52, a notable but controlled drop, demonstrating how the slip parameter cushions careless mistakes without erasing accumulated progress

2. Algorithmic Implementation in Flashcard Mode - Within the study space, the spaced repetition manages how the flashcards behave while BKT calculates the student's mastery of a concept. These flashcards can be made by the student or generated by the AI using the uploaded files. When a student opens a flashcard session, the system loads their current progress from the database. If it is a brand-new card set, the system initializes their score to the default prior $P(L_0) = 0.10$. During study, the student flips a card to reveal its definition and rates how well they remembered it by choosing Easy, Medium, or Hard.

Each rating maps to a binary observation and to the confidence-based guess probability established in Section 1.1, and it simultaneously sets the spaced-repetition interval at which the card will reappear. Choosing Easy means the student remembered the card easily; the system records a correct answer ($obs = 1$), applies the low-confidence guess rate $P(G_easy)$ so that the

mastery boost reflects genuine knowledge rather than luck, and schedules the card to reappear in seven days. This longer interval corresponds to a later review tier of the Leitner spaced-repetition system, which prescribes progressively longer intervals for well-recalled items to optimize long-term retention while minimizing unnecessary review (Durrani et al., 2024).

Choosing Medium means the student remembered the card but had to think about it; the system again records a correct answer $(obs = 1)$ but applies the higher medium-confidence guess rate $P(G_medium)$, yielding a smaller and more cautious mastery boost, and schedules the card to reappear in three days. This interval aligns with an intermediate review tier of the Leitner system, ensuring moderately recalled items are revisited sooner than well-recalled items to reinforce retention before significant forgetting occurs (Durrani et al., 2024).

Finally, choosing Hard means the student forgot the card; the system records an incorrect answer $(obs = 0)$, applies the hard-confidence guess rate $P(G_hard)$ from Section 1.1, and schedules the card to reappear in one day for immediate re-exposure. This shortest interval corresponds to the first review tier of the Leitner system, which prioritizes daily review for unrecalled or difficult items to maximize re-learning opportunities (Durrani et al., 2024). In every case the rating is sent directly to the BKT math engine, which applies the two-phase update from Section 1.1 to produce the student's new progress percentage $P(L_{+1})$ and saves it to the database. This hybrid setup gives students

personalized review times while the BKT engine accurately updates their knowledge statistics behind the scenes.

3. Algorithmic Implementation in Adaptive test Mode - Within the adaptive test mode, the Bayesian Knowledge Tracing model serves as the main engine for calibrating the difficulty of generated assessment sessions. When entering Test Mode, the student selects one or more custom question formats, such as multiple choice, identification, matching type, fill in the blank, or true or false. As the student completes their attempts during active testing, their performance is graded as a binary value (1 for correct or 0 for incorrect), and the BKT update pipeline calculates their dynamic mastery score on a per-question basis. Each attempt is evaluated using the question-type-specific guess probability defined in Section 1.1 together with the fixed slip probability $P(S) = 0.15$, so that the mastery gain from a correct answer is automatically scaled to how much chance the format allows. This real-time calculated score updates the student's concept-specific history in the central database.

When the student subsequently requests the generation of a new practice test session on the same topic, the system queries this historical BKT mastery probability to scale the conceptual difficulty of the newly generated items. The adaptive test generation engine utilizes specific mastery thresholds to adjust the generation prompt instructions sent to the large language model.

If the current running mastery probability is high, specifically at or above 0.85, the AI model is instructed to generate advanced-tier questions focusing on analytical scenarios, synthesis, and edge cases within the student's selected formats. If the mastery probability lies in the intermediate range between 0.50 and 0.85, the prompt parameters dictate intermediate-tier questions emphasizing conceptual comparisons, multi-concept recall, and practical applications. If the running mastery sits in the foundational range below 0.50, the generation instructions shift to produce simpler, highly supportive retrieval questions focusing on direct definitions and basic conceptual associations. This session-based adaptation allows students to retain complete control over their preferred test layouts while the system continuously adjusts the cognitive demands of the content to match their developmental needs.

4. Document Mastery and Gamification Calibration - While Bayesian Knowledge Tracing tracks individual concepts as isolated, high-precision knowledge components in the database, the system must provide a consolidated progress index for end-users. When a teacher of a student uploads a study document, the document ingestion engine parses and extracts a set of N distinct concept nodes. To render a high-readability overview on the student's dashboard, the application layer dynamically aggregates these independent probabilities using an arithmetic mean. The overall mastery rate M_doc for an uploaded document containing N concepts is calculated at any given timestamp using the most current running BKT metrics fetched from the database:

$$\text{EQ (4): } M_{doc} = \frac{\Sigma P(L_{current,i})}{N}$$

Where M_doc represents the overall document mastery percentage displayed as a progress bar on the user dashboard; N represents the total number of distinct concepts extracted from the uploaded document file; and $P(L_current,\ i)$ represents the real-time, running BKT mastery probability of the i-th concept stored under the student's profile in the database. Each concept's probability is calculated through the sequential, chronological application of the core BKT equations detailed in the BKT Mathematical Model. At the initial state (before any student interactions, where t = 0), the concept's probability is initialized to the baseline prior $P(L_0) = 0.10$. For each subsequent practice attempt, the system captures the binary response, calculates the intermediate conditional probability using the corresponding evidence formula, and executes the transition update to produce the updated probability state. The active value referenced in this aggregation is the most recently computed and saved $P(L_{+1})$ stored within the student's concept profile, ensuring the aggregate overview dynamically adapts to their live learning progress.

This aggregated score is directly utilized by the system's gamification module to calibrate the operational difficulty of generated games (crosswords, term matching, and timed trivia). When a game session is initialized, the system retrieves the current M_{doc} value from the database and applies it as a baseline parameter to the generative AI prompt. For example, if a student's document mastery sits at exactly 50%, the game engine utilizes this value to restrict the

vocabulary complexity, clue difficulty, and trivia response timers to align exactly with that 50% knowledge threshold. This integration protects students from cognitive frustration while keeping gamified learning tightly synchronized with their cognitive progress.

5. Evaluation in Free Recall - Within the study spaces, the Free Recall tab lets students type out everything they can remember about a lesson without looking at their notes. To grade this automatically, the system uses a method called Cosine Similarity which is calculated natively inside the database using the Supabase pgvector extension. This database-level calculation removes the need for complex mathematical scripts in the application's backend code by computing vector geometry directly on the cloud server.

When a student uploads their study materials, the application breaks the document into small pieces and converts those pieces into lists of numbers called high-dimensional vector embeddings that capture their exact meaning. When the student types out their recall summary, the system converts their typed text into a vector too.

Instead of running slow calculations on the backend application layer, the backend simply sends the student's vector to Supabase. The database natively calculates the mathematical angle between the student's typed vector $\vec{A}$ and the lesson's stored vector $\vec{B}$ using the Cosine Distance Operator <=>, where the final Cosine Similarity is found through the simple subtraction formula:

$$Cosine\ Similarity\ =\ 1\ -\ (A\ <=>\ B)$$

If the database returns a score close to 1, it means the student's written ideas point in the same directional meaning as the lesson. If the similarity score falls below a set quality threshold (such as 0.75), the system identifies that a gap exists. The system then compares the student's typed text directly against the lesson's key glossary terms to spot exactly which keywords were missing, displaying this feedback immediately to the student so they know what to review.

6. Feedback generation in Feynman Technique - The Feynman Technique helps students learn by explaining hard topics in simple, everyday language, as if they are teaching a 10-year-old child. To evaluate how well a student does this, the application guides a large language model (LLM) through a three-step evaluation pipeline.

The first step is the Jargon Check. The AI scans the student's written explanation and compares it against the main glossary list extracted during the document upload phase. If the student uses highly complex terms (like "photosynthesis" or "electromagnetism") without breaking them down simply, the AI flags those words and tells the student they need to explain them in simpler terms.

The second step is Analogy Verification. The AI checks if the student used a helpful comparison or everyday story (an analogy) to make the idea clear. For

example, when explaining electrical circuits, the AI checks if the student compared it to water flowing through pipes to make the concept easier to grasp.

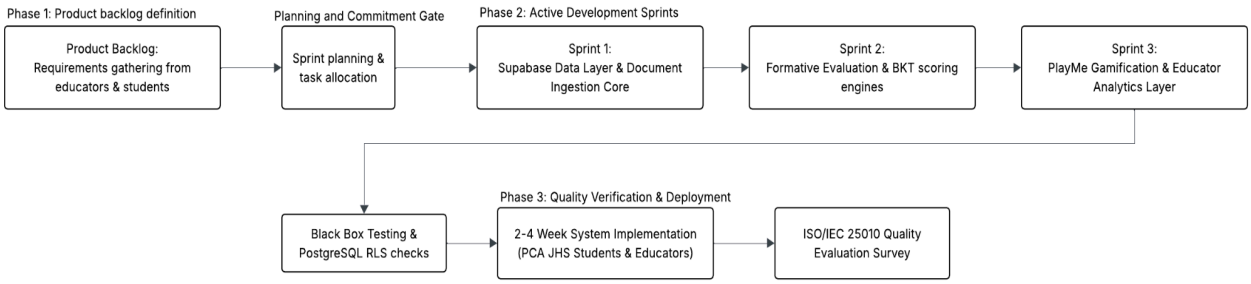
The third and final step is Finding the Gaps. The AI compares the student's simple explanation against the concept's true definition stored in the system to check whether anything is factually wrong, missing, or incomplete. Based on this comparison, it produces an accuracy score from 0 to 100 and a matching verdict of Correct, Needs Improvement, or Incorrect. The AI does not just hand back a grade. It also writes short, encouraging feedback that points out what the student got right, names any factual errors directly, and tells them specifically how to improve so they can close their own learning gaps.

SOFTWARE ENGINEERING PARADIGM

This study utilizes the Agile Scrum framework to guide the design, development, and implementation of LearnSync AI. Since the platform includes several interconnected features that require continuous testing and improvement, a flexible software development methodology is needed. Agile Scrum supports iterative development through short sprint cycles, allowing developers to continuously build, test, and refine the system while responding to changing requirements and user feedback (Hodson et al., 2024). Using Agile Scrum also enables the researchers to identify and resolve issues early in the development process, resulting in better software quality and reduced development risks

(Strode et al., 2022). This approach ensures that LearnSync AI is developed efficiently while meeting both user needs and system requirements.

Figure 2. *Agile SDLC*



The execution of the Agile Scrum framework across the development lifecycle is structured into the following sequential phases:

1. **Phase 1: Product Backlog** - The research team establishes the master list of software requirements from the perspective of two user roles: the Educator and the Student. The Educator must be able to upload their learning materials, select which JHS grade level can access the uploaded materials, and monitor student mastery through a class-level analytics dashboard. The Student must be able to access educator-uploaded materials and engage with four self-paced study options (adaptive flashcards, Free Recall with AI feedback, Feynman Technique explanations, and an AI chatbot), generate customized adaptive practice tests of various formats (multiple-choice, matching, fill-in-the-blank, identification, and true/false) that scale in difficulty based on their BKT

mastery state, and access a gamified review engine (crosswords, matching cards, and timed trivia) that adjusts difficulty using BKT scores.

2. **Planning and Commitment Stage** - The team conducts sprint planning sessions to break down backlog specifications into modular engineering tasks, estimate complexities, designate priorities, allocate tasks, and establish a sprint calendar. This ensures development cycles remain focused, preventing scope creep and prioritizing high-risk mathematical models and data security layers before active coding begins.

3. **Phase 2: Active Development Sprints -** The core coding lifecycle, structured into three iterative sprints:

    a. **Sprint 1: Supabase Data Layer & Document Ingestion Core** - Configures PostgreSQL tables on Supabase for educator profiles, student profiles, grade-level material access rules, and learning materials and embeddings records. Enforces Row-Level Security (RLS) policies to isolate each student's data under their authenticated user ID and restrict educators to mastery data within their selected grade level. Implements LLM-based text extraction scripts to parse educator-uploaded materials, extract concepts and definitions, and generate semantic vectors stored directly in the database.

    b. **Sprint 2: Formative Evaluation & BKT Scoring Engines** - Programs the four-parameter BKT scoring calculations to track

student mastery after each attempt (0 for incorrect, 1 for correct) and develops the adaptive test routing algorithm that adjusts question difficulty based on running mastery metrics. Establishes vector-based semantic evaluation for Free Recall similarity checks, integrates LLM prompt templates for Feynman explanations and chatbot queries, and logs activity timestamps to the Activity and Streak Logs database.

c. **Sprint 3: PlayMe Gamification & Educator Analytics Layer**: Codes the backtracking layout algorithm for crossword grids, trivia questions, and vocabulary matches using parsed glossaries and BKT scores as difficulty references. Builds the gamification and streak engine to validate badge criteria and manage daily streaks from a unified activity log. Programs the educator analytics dashboard to aggregate student mastery data at the concept and document level into a real-time grade-level monitoring interface. Compiles the backend API gateways with a responsive, dual-portal web frontend serving both Educator and Student interfaces.

4. **Phase 3: Quality Verification & Development** - The completed application undergoes Black Box Testing and PostgreSQL RLS checks to verify that data isolation operates correctly for both student-to-student and educator access boundaries. The application is then launched for a 2 to 4 week live implementation at Pasig Christian Academy, where educators upload materials through the Educator Portal while JHS students

independently use the platform to study their curriculum materials. At the conclusion, a structured ISO/IEC 25010 quality evaluation survey is administered using a 4-point Likert scale to measure Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability

DATA FLOW DIAGRAM

This section presents the Data Flow Diagrams (DFDs) for LearnSync AI. A DFD shows how information moves through the system, from the moment an educator uploads a learning material to the point where the system returns adaptive tests, feedback, and progress updates to students and mastery reports to educators. The Level 0 diagram gives a single high-level view of the whole system, and the Level 1 diagram breaks that system down into its main processes and data stores.

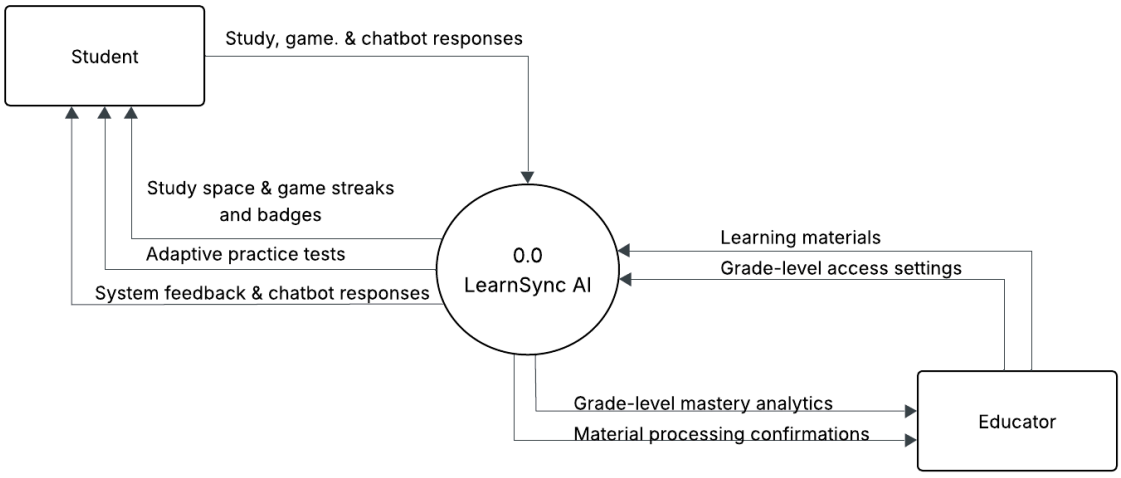
1. **Level 0: Context Diagram -** The context diagram represents the entire application as a single process, 0.0 LearnSync AI, with two external entities: the Educator and the Student.

   The Educator sends Learning materials (PDF documents) and Grade-level access settings (specifying which grade level can access the uploaded materials) into the system. The system returns Grade-level

mastery analytics (real-time student mastery progress reports) and Material processing confirmations (status responses confirming whether uploads were successfully parsed and made available).

The Student sends Study, game & chatbot responses (flashcard ratings, test answers, game interactions, and chatbot queries) into the system. The system returns three outputs: Study space & game streaks and badges (study hours, daily streaks, and earned badges), Adaptive practice tests (quizzes scaled to the student's mastery level), and System feedback & chatbot responses (AI-generated study guidance and chatbot replies).

Figure 3. *Level 0 Context Diagram*



2. **Level 1: Date flow diagram** - The Level 1 diagram contains 5 processes, 3 data stores (D1: Study Spaces, Learning Materials and Embeddings;

D2: Student Profiles and Mastery Track; D3: Activity and Streak Logs), and 2 external entities (Educator, Student).

Process 1.0 Material Upload and Distribution Control receives Learning materials and Grade-level access settings from the Educator, then stores the raw files and access configuration in D1 through the Raw files & materials access rules flow. When a Student opens their study space, the process checks D1 for matching access rules and returns the Authorized learning materials for that grade level. Process 1.0 also reads aggregated mastery data from D2 through the Grade-level mastery analytics flow and delivers it to the Educator for student progress monitoring.

Process 2.0 Document Parsing and Semantic Ingestion retrieves raw files from D1 through the Raw uploaded documents flow, extracts keywords, and converts the text into vector embeddings. It writes the results back to D1 through the Parsed text, keywords and vectors flow and sends a Material processing confirmation to the Educator.

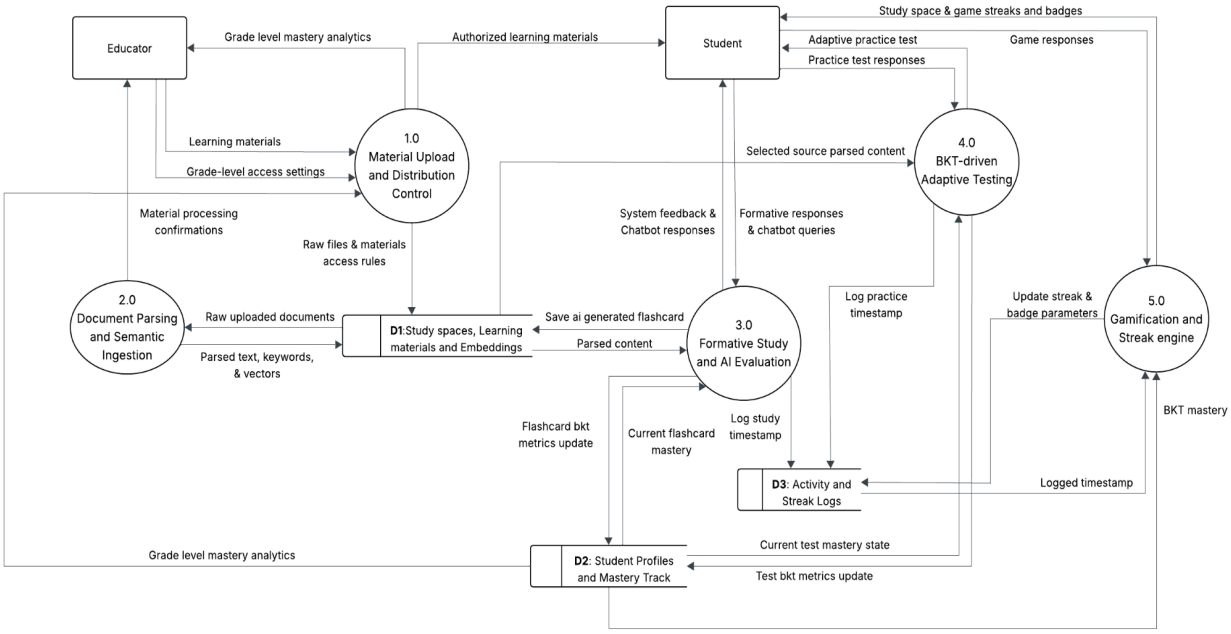
Process 3.0 Formative Study and AI Evaluation handles flashcards, the Feynman Technique, free recall, and the AI chatbot. It receives Formative responses & chatbot queries from the Student, reads Parsed Content from D1, and returns System feedback & chatbot responses. AI-generated flashcards are saved back to D1 through the Save AI-generated Flashcards flow. For mastery tracking, the process reads the

student's Current flashcard mastery from D2, applies the BKT calculation based on the flashcard rating, and writes the updated score back to D2 through the Flashcard BKT metrics update flow. It also logs each session to D3 through the Log study timestamp flow.

Process 4.0 BKT-driven Adaptive Testing receives Practice test responses from the Student, reads the Selected source parsed content from D1 and the Current test mastery state from D2, applies BKT to estimate mastery, and returns Adaptive practice tests matched to the student's level. Updated scores are written to D2 through the Test BKT metrics update flow, and session times are logged to D3 through the Log practice timestamp flow.

Process 5.0 Gamification and Streak Engine receives Game responses from the Student, reads the document-level BKT mastery score (M_doc) from D2 to calibrate game difficulty, and reads the Logged timestamp from D3 to check daily study activity. It updates streak and badge records in D3 through the Update streak & badge parameters flow and returns Study space & game streaks and badges to the Student.

Figure 4. *Level 1 Main Process*



TOOL TO TEST FOR ACCURACY

To ensure the reliability and correctness of LearnSync AI, the study employs a multi-layered validation strategy that evaluates both the algorithmic implementation and the functional behavior of the system. Since LearnSync AI does not utilize historical student interaction data, model training, or data-driven parameter estimation, the evaluation does not employ traditional machine learning metrics such as Confusion Matrix, Precision, Recall, F1-Score, or Mean Absolute Error. Instead, the study adopts a software engineering and implementation verification approach that validates whether the system's core modules, the Bayesian Knowledge Tracing (BKT) engine and the Generative AI

components, operate correctly according to their specified mathematical formulas and intended functional behavior.

1. **BKT Algorithm Implementation Verification**

The BKT module is evaluated through deterministic algorithm verification, confirming whether the system's programmatic implementation produces mastery estimates that are mathematically identical to hand-computed expected values. Since the study utilizes predefined default parameters from established educational data mining literature rather than parameters learned through Expectation Maximization, the objective is not to assess whether BKT accurately predicts real-world mastery, but to verify that the implemented code faithfully executes the standard four-parameter update equations without computational errors or logical defects.

The researchers design structured test cases, each consisting of a known initial mastery state, a specific sequence of binary responses (correct or incorrect), the default parameters $P(L_0) = 0.10$, $P(T) = 0.20$, $P(S) = 0.15$, and the context-sensitive P(G) values corresponding to the interaction mode, along with the expected mastery output manually computed using the BKT evidence and transition update formulas. These test cases cover consecutive correct responses, consecutive incorrect responses, alternating responses, and mixed sequences across Adaptive

Flashcard ratings (Easy, Medium, Hard) and Adaptive Test Mode question types (multiple choice, true or false, identification, fill in the blank, and matching type). Each test case is executed in the live application, and the system-generated mastery value is compared against the manually computed expected value.

Implementation correctness is quantified using the Implementation Correctness Rate (ICR):

$$\textit{EQ (5): } ICR = \frac{Number\ of\ Passed\ Test\ Cases}{Total\ Number\ of\ Test\ Cases} X\ 100$$

A test case is considered passed if the absolute difference between the system-generated and manually computed values falls within a tolerance of ± 0.01, accounting for floating-point arithmetic variations. An ICR of 100% indicates full and correct implementation. Any failed test case is documented with its expected value, actual output, and computed deviation for traceability.

This verification validates the correctness of the algorithm's implementation, not the predictive validity of the BKT model itself. The study confirms that given input parameters and observed responses, the system produces mastery estimates mathematically consistent with the standard BKT formulas (Bulut et al., 2023).

2. Generative AI Functional Verification

The Generative AI components, AI-generated flashcards, adaptive test questions, educational game content, chatbot responses, Feynman Technique feedback, and Free Recall evaluation feedback are evaluated through functional verification rather than model-level benchmarking. Since the study does not fine-tune or retrain the underlying large language model, the evaluation does not assess the AI model's internal accuracy. Instead, it verifies that outputs are contextually relevant, factually consistent with the uploaded learning resource, aligned with the intended BKT-driven difficulty level, and appropriate for the specified interaction mode.

The researchers upload a controlled learning resource with known content and systematically trigger each AI-powered feature. Each generated output is evaluated against four criteria:

Contextual Relevance assesses whether the generated content (flashcard terms and definitions, quiz questions and answers, game clues, chatbot responses, and feedback text) directly pertains to the topics, concepts, and terminology present in the uploaded learning resource. Outputs that reference information outside the scope of the uploaded material or introduce unrelated content are flagged as contextually irrelevant.

Factual Correctness assesses whether the generated content is factually accurate compared to the source material, including quiz answer keys, flashcard definitions, and chatbot responses.

Difficulty Alignment assesses whether the content reflects the appropriate difficulty tier based on the student's current BKT mastery level: foundational (below 0.50), intermediate (0.50–0.85), or advanced (at or above 0.85). This confirms that the adaptive calibration pipeline correctly passes the mastery parameter to the generative AI prompt.

Content Consistency assesses whether the AI produces stable, non-contradictory outputs across multiple generations for the same resource and mastery level. Minor phrasing variations are expected, but core factual content, terminology, and difficulty level should remain consistent.

Results are reported descriptively, documenting the total outputs evaluated, the number and percentage satisfying all criteria, and any notable deviation patterns.

3. Black-Box Functional Testing

The study employs Black-Box Functional Testing to validate the overall behavior of LearnSync AI as an integrated software system. Black-box testing evaluates external behavior based on specified functional requirements without examining internal source code or database structure.

The researchers design test cases corresponding to the platform's core functional requirements for both user roles. Educator-side test cases include PDF document upload and content extraction & grade-level access configuration for uploaded materials, and the class-level mastery analytics dashboard. Student-side test cases include authentication, access to educator-uploaded learning materials, flashcard generation and spaced repetition scheduling, adaptive test session generation and scoring, educational game generation and difficulty calibration, BKT mastery tracking and dashboard visualization, Free Recall submission and cosine similarity evaluation, Feynman Technique submission and AI feedback generation, AI chatbot interaction, and learning analytics display. Each test case specifies a precondition, an input action, and an expected output. The test is executed on the live deployed application, and results are documented in a structured log recording the feature tested, the expected result, the actual result, and the pass or fail status.

4. ISO/IEC 25010 Software Quality Model Survey

To evaluate overall software quality from the end user's perspective, the study employs the ISO/IEC 25010 Software Quality Model. The platform is evaluated across the eight quality characteristics defined by the standard: Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

A structured survey questionnaire aligned with the ISO/IEC 25010 characteristics is administered to the participants following the 2 to 4 week implementation period. The questionnaire utilizes a Likert scale to capture perceived quality across the eight characteristics. Responses are analyzed using descriptive statistics, including mean scores and standard deviations, to derive an overall quality assessment per characteristic. These results provide empirical evidence of the platform's perceived quality and acceptability, complementing the technical verification results from the BKT algorithm verification, Generative AI functional verification, and black-box functional testing.

POPULATION AND SAMPLING

The target population of this study consists of Junior High School (JHS) students (Grades 7 - 10) of Pasig Christian Academy for the Academic Year 2026–2027. As the intended end-users of LearnSync AI, these students will participate in the implementation and evaluation of the proposed system. Junior

High School students are the target population as they are in a stage of developing independent learning habits, self-regulated learning skills, and academic readiness, making them suitable participants for evaluating the effectiveness of the proposed platform.

The study will employ Proportionate Stratified Random Sampling to select 50 Junior High School students from the total Junior High School population of 100 students. Respondents will first be grouped according to their respective grade levels (Grades 7 - 10). The number of participants selected from each grade level will be proportional to its population size, as presented in Table 1. Within each grade level, respondents will be selected through simple random sampling. This sampling technique ensures that all grade levels are fairly represented while minimizing selection bias.

Table 1. *Population and Sample Distribution*

Grade Level	**Population (N)**	**Sample (n)**
Grade 7	18	9
Grade 8	29	15
Grade 9	34	17
Grade 10	19	9
Total	**100**	**50**

The sample size for each grade level was determined using the proportionate allocation formula:

$$\textit{EQ (6):}\ n_h = \frac{N_h}{N}\, x\, n$$

Where:

- n_h = Sample size for each grade level
- N_h = Population of each grade level
- N = Total Population
- n = Total Sample Size

Although the respondents will be selected from different grade levels, the study does not intend to compare learning outcomes across Grades 7, 8, 9, and 10. Instead, all selected participants will be analyzed as a single study population for the implementation and evaluation of LearnSync AI. During the implementation period, the respondents will utilize the platform's core features, including Adaptive Flashcards, Adaptive Test Mode, AI-assisted study strategies, AI-generated educational games, the AI-powered learning assistant, and learning analytics. Their individual learning progress and knowledge mastery will be monitored using Bayesian Knowledge Tracing (BKT) to measure changes in mastery throughout the implementation period. Consequently, the analysis will

focus on within-student mastery progression throughout the implementation period rather than differences between grade-level groups.

EVALUATION INSTRUMENT

The primary evaluation instruments used in this study consist of researcher-developed questionnaires and system-generated learning analytics designed to assess LearnSync AI in terms of software quality, user acceptability, and students' knowledge mastery progression. The evaluation consists of three components. First, the platform automatically records students' knowledge mastery progression using the Bayesian Knowledge Tracing (BKT) model throughout the implementation period. Second, Junior High School students and educators evaluate the acceptability of LearnSync AI using questionnaires based on the ISO/IEC 25010 Software Quality Model. Finally, qualitative feedback is collected to gather participants' comments, observations, and recommendations for future system improvement.

The ISO/IEC 25010 questionnaire evaluates the platform using the applicable software quality characteristics of Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. Students evaluate the student-facing functionalities of LearnSync AI, while educators evaluate only the functionalities accessible through the educator module, including learning resource upload and management, grade-level assignment of learning resources, and student learning statistics.

- **Quantitative Instrument:** The quantitative instrument consists of structured close-ended statements designed to evaluate the quality and acceptability of LearnSync AI. Respondents rate each statement using a 4-point Likert scale, which eliminates the neutral option to encourage a more definitive evaluation of the system:

    - **4 – Strongly Agree (SA) / Excellent:** The evaluated functionality consistently performs as intended and fully satisfies the expected quality standards.
    - **3 – Agree (A) / Good:** The evaluated functionality consistently performs as intended and fully satisfies the expected quality standards.
    - **2 – Disagree (D) / Fair:** The evaluated functionality consistently performs as intended and fully satisfies the expected quality standards.
    - **1 – Strongly Disagree (SD) / Poor:** The evaluated functionality consistently performs as intended and fully satisfies the expected quality standards.

- **Qualitative Feedback:** The final section consists of open-ended questions that allow both students and educators to provide comments, observations, suggestions, and recommendations regarding LearnSync AI. This feedback enables respondents to describe their experiences using the platform, identify usability or functional concerns, and recommend possible improvements that may not be adequately captured through the quantitative evaluation.

DATA COLLECTION METHOD

Data for this study will be collected through four phases: requirement analysis, pre-implementation, system implementation, and post-implementation evaluation. These phases ensure that the identified needs of Pasig Christian Academy are translated into the design, development, implementation, and evaluation of LearnSync AI.

During the requirement analysis phase, the researchers will conduct interviews and questionnaires with the school principal, educators, and Junior High School students of Pasig Christian Academy to identify the current independent learning process, existing challenges, and the functional and non-functional requirements of the proposed system. The information gathered during this phase will serve as the basis for the design and development of LearnSync AI.

During the pre-implementation phase, the researchers will obtain permission from the school administration and secure informed consent from all participants. The objectives, procedures, and duration of the study will be explained, and participants will receive an orientation on how to access and use LearnSync AI.

During the implementation phase, 50 selected Junior High School students will use LearnSync AI for two to four weeks as part of their independent learning outside regular classroom instruction. The implementation will initially focus on the subject areas identified through the requirement analysis as the most challenging for each participating grade level. This selection is intended only to provide the initial learning content for system evaluation and does not limit LearnSync AI to a specific subject area.

Students will utilize the platform's core features, including Adaptive Flashcards, Adaptive Test Mode, the AI Learning Assistant, AI-assisted study strategies through the Feynman Technique and Free Recall, AI-generated educational games, and learning analytics. Throughout the implementation, the system will automatically record each student's assessment results, study activity, and Bayesian Knowledge Tracing (BKT) mastery estimates. The researchers will monitor the implementation and provide technical assistance whenever necessary.

During the post-implementation phase, Junior High School students and educators will complete the ISO/IEC 25010 Software Quality Evaluation

Questionnaire through Google Forms. Students will evaluate the student-facing functionalities of LearnSync AI, while educators will evaluate only the functionalities available through the educator module. An open-ended questionnaire will also be administered to gather comments, suggestions, and recommendations for future system improvements.

The responses collected through Google Forms will be automatically recorded for statistical analysis, while the BKT mastery records generated by LearnSync AI will be used to evaluate each student's knowledge mastery progression throughout the implementation period.

ETHICAL CONSIDERATIONS

This study will be conducted in accordance with established ethical research principles to protect the rights, privacy, and welfare of all participants. Prior to the implementation of LearnSync AI, permission will be obtained from the administration of Pasig Christian Academy, and informed consent will be secured from participating Junior High School students, educators, and parents or guardians, when applicable. Participants will be informed of the study's purpose, procedures, duration, and their right to withdraw from the study at any time without penalty. Participation will be entirely voluntary.

The study will comply with the Data Privacy Act of 2012 (Republic Act No. 10173) by collecting only the information necessary for system implementation and research. Personal information, academic records, and system-generated

learning data, including Bayesian Knowledge Tracing (BKT) mastery records, will be treated with strict confidentiality and used solely for academic research and system evaluation. No personally identifiable information will be disclosed in any reports, presentations, or publications.

To maintain academic integrity, only authenticated educators will be authorized to upload learning resources used by the platform. LearnSync AI incorporates appropriate security measures to protect participant data, while access to learning analytics and personal learning records is restricted to the respective student and authorized educators. All collected information will be used exclusively for academic research and system evaluation and will not be shared with unauthorized third parties.

SUMMARY

This chapter presented the research methodology used in the development and evaluation of LearnSync AI, an AI-assisted learning platform designed to support the independent learning of Junior High School students with an accompanying Educator Portal for material management and student mastery monitoring. It discussed the research design, system architecture, software engineering methodology, research locale, population and sampling, evaluation instrument, data collection procedures, system validation methods, statistical treatment of data, and ethical considerations.

The study employed a developmental research design using quantitative methods to support the analysis, design, development, implementation, and evaluation of the proposed system. LearnSync AI was developed using the Agile Scrum software development methodology and integrates the classical Bayesian Knowledge Tracing (BKT) model to estimate students' mastery levels. Educators upload and manage the learning materials that students access through the platform. Adaptive Flashcards and Adaptive Test Mode update students' estimated mastery, while AI-assisted learning strategies, AI-generated educational games, and personalized learning activities utilize the current mastery estimates to support independent learning. The Educator Portal provides a grade-level analytics dashboard that enables teachers to monitor student mastery progress in real time.

The study will be conducted at Pasig Christian Academy involving Junior High School students and their assigned educators, with student respondents selected through Proportionate Stratified Random Sampling. The system will be implemented over a period of 2 to 4 weeks, during which educators will upload learning materials through the Educator Portal while students use the platform as part of their independent learning activities. Data will be collected using researcher-developed questionnaires to evaluate the software quality of LearnSync AI based on the ISO/IEC 25010 Software Quality Model, covering Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability.

In addition, the correctness of the Bayesian Knowledge Tracing implementation will be verified by comparing the system-generated mastery values with manually computed expected values using predefined test cases. The Generative AI components will also undergo functional verification to ensure that the generated outputs are contextually relevant, factually consistent, and aligned with the uploaded learning resources. Black-box functional testing will likewise be conducted to verify the expected behavior of the platform's major features across both the Educator and Student portals.

Finally, the collected data will be analyzed using appropriate statistical tools, while ethical principles, including compliance with the Data Privacy Act of 2012 (Republic Act No. 10173), will be observed throughout the study. The methodology presented in this chapter provides a structured framework for evaluating the implementation and software quality of LearnSync AI as an AI-assisted platform that supports the independent learning of Junior High School students under educator-supervised content management.

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